US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250279240
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	WATANABE; Hiroto

MULTILAYER CERAMIC ELECTRONIC COMPONENT

Abstract

A multilayer ceramic electronic component includes a multilayer body and a pair of external electrodes at both ends of the multilayer body. The external electrodes include a main surface-side external electrode including a recess recessed towards a multilayer body side in a cross-sectional view along a lamination direction and a length direction of the multilayer body, and peripheral regions adjacent to the recess in the length direction. A surface roughness of the recess is rougher than a surface roughness of the peripheral regions.

Inventors: **WATANABE; Hiroto (Nagaokakyo-shi, JP)**

Applicant: **Murata Manufacturing Co., Ltd.** (Nagaokakyo-shi, JP)

Family ID: **96816528**

Appl. No.: **18/619241**

Filed: **March 28, 2024**

Foreign Application Priority Data

JP	2024-030487	Feb. 29, 2024

Publication Classification

Int. Cl.: **H01G4/232** (20060101); **H01G4/12** (20060101); **H01G4/30** (20060101)

U.S. Cl.:

CPC **H01G4/232** (20130101); **H01G4/1245** (20130101); **H01G4/30** (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
[0001] This application is based on and claims the benefit of priority from Japanese Patent Application No. 2024-030487, filed on Feb. 29, 2024, the contents of which are incorporated herein by reference.
BACKGROUND OF THE INVENTION
1. Field of the Invention
[0002] The present invention relates to multilayer ceramic electronic components.
2. Description of the Related Art
[0003] Conventionally, multilayer ceramic capacitors as multilayer ceramic electronic components are known. In general, a multilayer ceramic capacitor includes a multilayer body, in which dielectric layers and internal electrode layers are alternately stacked, and external electrodes are provided on both end surfaces of the multilayer body. For example, Japanese Unexamined Patent Application, Publication No. 2003-243249 discloses a multilayer ceramic capacitor with the aforementioned structure, in which the external electrodes include a base electrode layer formed by firing.
[0004] In this type of multilayer ceramic capacitor, when mounted on a board, the bending stress occurring in the external electrodes is transmitted to the multilayer body, leading to concerns that cracks or the like may occur in the multilayer body. Therefore, there is a demand for a multilayer ceramic capacitor with improved flexural resistance.
SUMMARY OF THE INVENTION
[0005] Example embodiments of the present invention provide multilayer ceramic electronic components each with improved flexural resistance.
[0006] A multilayer ceramic electronic component according to an example embodiment of the present invention includes a multilayer body including a plurality of ceramic layers and a plurality of inner conductive layers stacked alternately in a lamination direction, first and second main surfaces on opposite sides in the lamination direction, first and second end surfaces on opposite sides in a length direction orthogonal or substantially orthogonal to the lamination direction, and first and second lateral surfaces on opposite sides in a width direction orthogonal or substantially orthogonal to both the lamination direction and the length direction, and a pair of external electrodes spaced apart from each other at both ends of the multilayer body in the length direction. The inner conductive layers include a first inner conductive layer extending to the first end surface, and a second inner conductive layer extending to the second end surface.
[0007] The external electrodes include a main surface-side external electrode on at least one of the first main surface or the second main surface. The main surface-side external electrode includes a recess recessed towards a multilayer body side in a cross-sectional view along the lamination direction and the length direction, and a peripheral region adjacent to the recess in the length direction. A surface roughness of the recess is rougher than a surface roughness of the peripheral region.
[0008] Example embodiments of the present invention provide multilayer ceramic electronic components each with improved flexural resistance.
[0009] The above and other elements, features, steps, characteristics s and advantages of the present invention will become more apparent from the following detailed description of the example embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0010] FIG. **1** is a perspective view of a multilayer ceramic capacitor according to an example embodiment of the present invention.

[0011] FIG. **2** is a cross-sectional view taken along the line II-II of FIG. **1**.
[0012] FIG. **3** is a cross-sectional view taken along the line III-III of FIG. **2**.
[0013] FIG. **4**A is a cross-sectional view taken along the line IVA-IVA of FIG. **2**.
[0014] FIG. **4**B is a cross-sectional view taken along the line IVB-IVB of FIG. **2**.
[0015] FIG. **5**A is an enlarged view of a portion indicated by VA in FIG. **2**, illustrating a cross section of a first main surface-side external electrode.
[0016] FIG. **5**B is a view corresponding to FIG. **5**A, illustrating a cross section of the first main surface-side external electrode.
[0017] FIG. **6**A is a diagram illustrating a method of manufacturing a multilayer ceramic capacitor according to an example embodiment of the present invention, illustrating a first step of forming external electrodes on the multilayer body.
[0018] FIG. **6**B is a diagram illustrating a method of manufacturing a multilayer ceramic capacitor to an example embodiment of the present invention, illustrating a second step of forming external electrodes on the multilayer body.
[0019] FIG. **6**C is a diagram illustrating a method of manufacturing a multilayer ceramic capacitor according to an example embodiment of the present invention, illustrating a third step of forming external electrodes on the multilayer body.
[0020] FIG. **7**A illustrates a multilayer ceramic capacitor according to an example embodiment of the present invention with a two-portion structure.
[0021] FIG. **7**B illustrates a multilayer ceramic capacitor according to an example embodiment of the present invention with a three-portion structure.
[0022] FIG. **7**C illustrates a multilayer ceramic capacitor according to an example embodiment of the present invention with a four-portion structure.

DETAILED DESCRIPTION OF THE EXAMPLE EMBODIMENTS

[0023] Example embodiments of the present invention will be described in detail below with reference to the drawings.
[0024] A multilayer ceramic capacitor **1** as a multilayer ceramic electronic component according to an example embodiment of the present disclosure will be described with reference to the drawings. FIG. **1** is a perspective view of the multilayer ceramic capacitor **1** according to an example embodiment of the present invention. FIG. **2** is a cross-sectional view taken along the line II-II of FIG. **1**. FIG. **3** is a cross-sectional view taken along the line III-III of FIG. **2**. FIG. **4**A is a cross-sectional view taken along the line IVA-IVA of FIG. **2**. FIG. **4**B is a cross-sectional view taken along the line IVB-IVB of FIG. **2**.
[0025] As illustrated in FIG. **1**, the multilayer ceramic capacitor **1** according to the present example embodiment has a rectangular or substantially rectangular parallelepiped shape. The multilayer ceramic capacitor **1** includes a multilayer body **10** having a rectangular or substantially rectangular parallelepiped shape, and a pair of external electrodes **40** spaced apart from each other at both ends of the multilayer body **10**.
[0026] In FIG. **1**, the arrow T indicates the lamination direction of the multilayer ceramic capacitor **1** and the multilayer body **10**. The lamination direction T is also the thickness direction and the height direction of the multilayer ceramic capacitor **1** and the multilayer body **10**. In FIG. **1**, the arrow L indicates the length direction orthogonal or substantially orthogonal to the lamination direction T of the multilayer ceramic capacitor **1** and the multilayer body **10**. In FIG. **1**, the arrow W indicates the width direction orthogonal or substantially orthogonal to both the lamination direction T and the length direction L of the multilayer ceramic capacitor **1** and the multilayer body **10**. The pair of external electrodes **40** are provided at ends of the multilayer body **10** in the length direction L, respectively.
[0027] FIGS. **1** to **4**B illustrate an XYZ orthogonal coordinate system. The length direction L of the multilayer ceramic capacitor **1** and the multilayer body **10** corresponds to the X direction. The width direction W of the multilayer ceramic capacitor **1** and the multilayer body **10** corresponds to

the Y direction. The lamination direction T of the multilayer ceramic capacitor **1** and the multilayer body **10** corresponds to the Z direction. The cross section illustrated in FIG. **2** is also referred to as an LT cross section. The cross section illustrated in FIG. **3** is also referred to as a WT cross section. The cross section illustrated in FIGS. **4**A and **4**B is also referred to as an LW cross section.

[0028] As illustrated in FIGS. **1** to **4**B, the multilayer body **10** includes a first main surface TS**1** and a second main surface TS**2** on opposite sides in the lamination direction T, a first end surface LS**1** and a second end surface LS**2** on opposite sides in the length direction L orthogonal or substantially orthogonal to the lamination direction T, and a first lateral surface WS**1** and a second lateral surface WS**2** on opposite sides in the width direction W orthogonal or substantially orthogonal to both the lamination direction T and the length direction L.

[0029] As illustrated in FIG. **1**, the multilayer body **10** has a rectangular or substantially rectangular parallelepiped shape. The dimension in the length direction L of the multilayer body **10** is not necessarily longer than the dimension in the width direction W. The corners and edges of the multilayer body **10** are preferably rounded. The corners are where three faces of the multilayer body intersect, and the edges are where two faces of the multilayer body intersect. The surfaces of the multilayer body **10** may include irregularities in an entirety or in a portion thereof.

[0030] The dimensions of the multilayer body **10** are not particularly limited. However, the dimension of the multilayer body **10** in the length direction L, denoted as the L dimension, is preferably, for example, between about 0.2 mm and about 10 mm inclusive. The dimension of the multilayer body **10** in the lamination direction T, denoted as the T dimension, is preferably, for example, between about 0.05 mm and about 10 mm inclusive. The dimension of the multilayer body **10** in the width direction W, denoted as the W dimension, is preferably, for example, between about 0.1 mm and about 10 mm inclusive.

[0031] As illustrated in FIGS. **2** and **3**, the multilayer body **10** includes an inner layer portion **11**, and a first main surface-side outer layer portion **12** and a second main surface-side outer layer portion **13** which sandwich the inner layer portion **11** in the lamination direction T.

[0032] The inner layer portion **11** includes a plurality of dielectric layers **20** as a plurality of ceramic layers, and a plurality of internal electrode layers **30** as a plurality of inner conductive layers, both of which are stacked alternately in the lamination direction T. The internal electrode layers **30** included in the inner layer portion **11** range from an internal electrode layer **30** closest to the first main surface TS**1** to another internal electrode layer **30** closest to the second main surface TS**2**, in the lamination direction T. In the inner layer portion **11**, the plurality of internal electrode layers **30** face each other via the dielectric layers **20**. The inner layer portion **11** defines and functions to generate capacitance, and operates as a capacitor.

[0033] The dielectric layers **20** are made of dielectric materials. The dielectric materials may be dielectric ceramics containing components such as, for example, $BaTiO_3$, $CaTiO_3$, $SrTiO_3$, or $CaZrO_3$. In addition to these main components, the dielectric materials may include accessory components such as, for example, Mn compounds, Fe compounds, Cr compounds, Co compounds, or Ni compounds. The dielectric materials preferably include, for example, $BaTiO_3$ as the main component.

[0034] The thickness of the dielectric layers **20** is preferably, for example, between about 0.2 μm and about 15 μm inclusive. The number of dielectric layers **20** to be stacked (laminated) is preferably, for example, between 10 and 1200 inclusive. The number of dielectric layers **20** is the total of the number of dielectric layers **20** in the inner layer portion **11**, and the number of the dielectric layers **20** in the first main surface-side outer layer portion **12** and the second main surface-side outer layer portion **13**.

[0035] The plurality of internal electrode layers **30** include a plurality of first internal electrode layers **31** as a plurality of first inner conductive layers, and a plurality of second internal electrode layers **32** as a plurality of second inner conductive layers. The first internal electrode layers **31** and the second internal electrode layers **32** are alternately provided in the lamination direction T with the dielectric layers **20** interposed therebetween. The first internal electrode layers **31** extend to the first end surface LS**1**. The second internal electrode layers **32** extend to the second end surface LS**2**. Hereinafter, when there is no need to distinguish between the first internal electrode layers **31** and the second internal electrode layers **32** for description, the first internal electrode layers **31** and the second internal electrode layers **32** may collectively be referred to as the internal electrode layers **30**.

[0036] As illustrated in FIG. **4**A, the first internal electrode

[0037] layer **31** includes a first counter portion **31**A and a first extension portion **31**B. The first counter portion **31**A is a region facing the second internal electrode layer **32** across the dielectric layer **20** and is provided inside the multilayer body **10**. The first extension portion **31**B is a portion extending from the first counter portion **31**A to the first end surface LS**1** and exposed at the first end surface LS**1**.

[0038] As illustrated in FIG. **4**B, the second internal electrode layer **32** includes a second counter portion **32**A and a second extension portion **32**B. The second counter portion **32**A is a region facing the first internal electrode layer **31** across the dielectric layer **20** and is provided inside the multilayer body **10**. The second extension portion **32**B is a portion extending from the second counter portion **32**A to the second end surface LS**2** and exposed at the second end surface LS**2**.

[0039] In the present example embodiment, the first counter portion **31**A and the second counter portion **32**A face each other across the dielectric layer **20**, so as to generate a capacitance and provide the characteristics of the capacitor.

[0040] The shapes of the first counter portion **31**A and the second counter portion **32**A are not particularly limited but are preferably rectangular or substantially rectangular. However, the corners of the rectangular-shaped portions may be rounded or extend diagonally. The shapes of the first extension portion **31**B and the second extension portion **32**B are not particularly limited but are preferably rectangular or substantially rectangular. However, the corners of the rectangular-shaped portions may be rounded or extend diagonally.

[0041] Both of the dimensions of the first counter portion **31**A and the first extension portion **31**B in the width direction W may be the same or substantially the same, or one of the dimensions may be smaller. Both of the dimensions of the second counter portion **32**A and the second extension portion **32**B in the width direction W may be the same or substantially the same, or one of the dimensions may be smaller.

[0042] The first internal electrode layers **31** and the second internal electrode layers **32** are made of appropriate conductive materials such as, for example, metals like Ni, Cu, Ag, Pd, Au, or alloys including at least one of these metals. When using an alloy, for example, the first internal electrode layers **31** and the second internal electrode layers **32** may be made of Ag—Pd alloy, for example.

[0043] The thickness of the first internal electrode layers **31** and the second internal electrode layers **32** is preferably, for example, between about 0.2 μm and about 2.0 μm inclusive. The total number of the first internal electrode layers **31** and the second internal electrode layers **32** is preferably, for example, between 10 and 1000 inclusive.

[0044] As illustrated in FIGS. **2** and **3**, the first main surface-side outer layer portion **12** is provided to the first main surface TS**1** side of the multilayer body **10**. The first main surface-side outer layer portion **12** is a collective portion including the plurality of dielectric layers **20** between the first main surface TS**1** and the internal electrode layer **30** closest to the first main surface TS**1**. On the other hand, the second main surface-side outer layer portion **13** is provided to the second main surface TS**2** side of the multilayer body **10**. The second main surface-side outer layer portion **13** is a collective portion including the plurality of dielectric layers **20** between the second main surface TS**2** and the internal electrode layer **30** closest to the second main surface TS**2**. The dielectric layers **20** used for the first main surface-side outer layer portion **12** and the second main surface-side outer layer portion **13** may be the same as the dielectric layers **20** used for the inner layer portion **11**.

[0045] The multilayer body **10** includes a counter electrode portion **11**E. The counter electrode portion **11**E is a portion where the first counter portion **31**A of the first internal electrode layer **31** faces the second counter portion **32**A of the second internal electrode layer **32**. The counter electrode portion **11**E is a portion of the inner layer portion **11**. FIGS. **4**A and **4**B illustrate the range of the counter electrode portion **11**E in the width direction W and the length direction L. The counter electrode portion **11**E is also referred to as the capacitor active portion.
[0046] The multilayer body **10** includes lateral surface-side
[0047] outer layer portions. The lateral surface-side outer layer portions include a first lateral surface-side outer layer portion WG**1** and a second lateral surface-side outer layer portion WG**2**. The first lateral surface-side outer layer portion WG**1** is a portion including the dielectric layers **20** between the counter electrode portion **11**E and the first lateral surface WS**1**. The second lateral surface-side outer layer portion WG**2** is a portion including the dielectric layers **20** between the counter electrode portion **11**E and the second lateral surface WS**2**. FIGS. **3**, **4**A, and **4**B illustrate the range of the first lateral surface-side outer layer portion WG**1** and the second lateral surface-side outer layer portion WG**2** in the width direction W. The lateral surface-side outer layer portions are also referred to as a W-gap or side gap.
[0048] The multilayer body **10** includes end surface-side outer layer portions. The end surface-side outer layer portions include a first end surface-side outer layer portion LG**1** and a second end surface-side outer layer portion LG**2**. The first end surface-side outer layer portion LG**1** is a portion including the dielectric layers **20** and the first extension portion **31**B between the counter electrode portion **11**E and the first end surface LS**1**. In other words, the first end surface-side outer layer portion LG**1** is a collective portion including a portion of the plurality of dielectric layers **20** on the first end surface LS**1** side and the plurality of first extension portions **31**B. Similarly, the second end surface-side outer layer portion LG**2** is a portion including the dielectric layers **20** and the second extension portion **32**B between the counter electrode portion **11**E and the second end surface LS**2**. In other words, the second end surface-side outer layer portion LG**2** is a collective portion including a portion of the plurality of dielectric layers **20** on the second end surface LS**2** side and the plurality of second extension portions **32**B. FIGS. **2**, **4**A, and **4**B illustrate the range of the first end surface-side outer layer portion LG**1** and the second end surface-side outer layer portion LG**2** in the length direction L. The end surface-side outer layer portions are also referred to as an L-gap or end gap.
[0049] As illustrated in FIGS. **1** and **2**, the external electrodes **40** include a first external electrode **40**A provided to the first end surface LS**1** side of the multilayer body **10**, and a second external electrode **40**B provided to the second end surface LS**2** side of the multilayer body **10**.
[0050] The basic structure of the first external electrode **40**A and the second external electrode **40**B is the same or substantially the same. The shape of the first external electrode **40**A and the second external electrode **40**B is plane-symmetrical or substantially plane-symmetrical with respect to the WT cross section at the center in the length direction L of the multilayer ceramic capacitor **1**. Therefore, when there is no need to distinguish between the first external electrode **40**A and the second external electrode **40**B for description, the first external electrode **40**A and the second external electrode **40**B may be collectively referred to as the external electrodes **40**.
[0051] The first external electrode **40**A is provided on the first end surface LS**1**. The first external electrode **40**A is in contact with the first extension portions **31**B of the plurality of first internal electrode layers **31** exposed at the first end surface LS**1**. Consequently, the first external electrode **40**A is electrically connected to the plurality of first internal electrode layers **31**. The first external electrode **40**A may also be provided on a portion of the first main surface TS**1**, a portion of the second main surface TS**2**, a portion of the first lateral surface WS**1**, and a portion of the second lateral surface WS**2**. In the present example embodiment, the first external electrode **40**A extends from the first end surface LS**1** to a portion of the first main surface TS**1**, a portion of the second main surface TS**2**, a portion of the first lateral surface WS**1**, and a portion of the second lateral

surface WS**2**.

[0052] The second external electrode **40**B is provided on the second end surface LS**2**. The second external electrode **40**B is in contact with each of the second extension portions **32**B of the plurality of second internal electrode layers **32** exposed at the second end surface LS**2**. Consequently, the second external electrode **40**B is electrically connected to the plurality of second internal electrode layers **32**. The second external electrode **40**B may be provided on a portion of the first main surface TS**1**, a portion of the second main surface TS**2**, a portion of the first lateral surface WS**1**, and a portion of the second lateral surface WS**2**. In the present example embodiment, the second external electrode **40**B extends from the second end surface LS**2** to a portion of the first main surface TS**1**, a portion of the second main surface TS**2**, a portion of the first lateral surface WS**1**, and a portion of the second lateral surface WS**2**.

[0053] As described above, within the multilayer body **10**, the first counter portion **31**A of the first internal electrode layer **31** faces the second counter portion **32**A of the second internal electrode layer **32** via the dielectric layer **20**, so as to generate capacitance. Therefore, capacitor characteristics a provided between the first external electrode **40**A connected to the first internal electrode layer **31** and the second external electrode **40**B connected to the second internal electrode layer **32**.

[0054] As illustrated in FIGS. **2**, **4**A, and **4**B, the first external electrode **40**A includes a first base electrode layer **50**A and a first plated layer **60**A provided on the first base electrode layer **50**A. The second external electrode **40**B includes a second base electrode layer **50**B and a second plated layer **60**B provided on the second base electrode layer **50**B.

[0055] The first base electrode layer **50**A is provided on the first end surface LS**1**. The first base electrode layer **50**A is connected to the first extension portions **31**B of the plurality of first internal electrode layers **31** exposed at the first end surface LS**1**. In the present example embodiment, the first base electrode layer **50**A extends from the first end surface LS**1** to a portion of the first main surface TS**1**, a portion of the second main surface TS**2**, a portion of the first lateral surface WS**1**, and a portion of the second lateral surface WS**2**.

[0056] The second base electrode layer **50**B is provided on the second end surface LS**2**. The second base electrode layer **50**B is in contact with the second extension portions **32**B of the plurality of second internal electrode layers **32** exposed at the second end surface LS**2**. In the present example embodiment, the second base electrode layer **50**B extends from the second end surface LS**2** to a portion of the first main surface TS**1**, a portion of the second main surface TS**2**, a portion of the first lateral surface WS**1**, and a portion of the second lateral surface WS**2**.

[0057] The first base electrode layer **50**A and the second base electrode layer **50**B of the present example embodiment are fired layers. The fired layer preferably include, for example, a metal component and either a glass component or a ceramic component, or both. The metal component may include, for example, at least one of Cu, Ni, Ag, Pd, Ag—Pd alloy, or Au. The glass component may include, for example, at least one of B, Si, Ba, Mg, Al, or Li. The ceramic component may use the same ceramic material as the dielectric layer **20** or a different type of ceramic material. Examples of the ceramic component include, for example, at least one of BaTiO.sub.3, CaTiO.sub.3, (Ba, Ca) TiO.sub.3, SrTiO.sub.3, or CaZrO.sub.3.

[0058] The fired layer is formed by applying a conductive paste including glass and metal to the multilayer body **10** followed by firing. The fired layer can be formed by simultaneously firing a pre-firing multilayer chip, which is a material of the multilayer body **10** including a plurality of internal electrodes and dielectric layers, and the conductive paste applied to the multilayer chip. Alternatively, the fired layer can be formed by obtaining the multilayer body **10** by firing the multilayer chip and then applying the conductive paste to the multilayer body **10** followed by firing. In the case as described above, the fired layer is preferably formed by firing a mixture including ceramic material instead of a glass component. In this case, as the ceramic material to be added, using a ceramic material the same as or similar to the dielectric layer **20** is preferable. The

fired layer may include a plurality of layers.
[0059] The thickness of the first base electrode layer **50**A provided on the first end surface LS**1** in the length direction L is, for example, preferably between about 2 μm and about 220 μm inclusive at the center of the first base electrode layer **50**A in the lamination direction T and the width direction W.
[0060] The thickness of the second base electrode layer **50**B provided on the second end surface LS**2** in the length direction L is, for example, preferably between about 2 μm and about 220 μm inclusive at the center of the second base electrode layer **50**B in the lamination direction T and the width direction W.
[0061] In cases where the first base electrode layer **50**A is also provided on a portion of at least one of the first main surface TS**1** or the second main surface TS**2**, the thickness of the first base electrode layer **50**A provided in this portion in the lamination direction T is, for example, preferably between about 3 μm and about 40 μm inclusive at the center of the first base electrode layer **50**A provided in this portion in the length direction L and the width direction W.
[0062] In cases where the first base electrode layer **50**A is also provided on a portion of at least one of the first lateral surface WS**1** or the second lateral surface WS**2**, the thickness of the first base electrode layer **50**A provided in this portion in the width direction W is, for example, preferably between about 3 μm and about 40 μm inclusive at the center of the first base electrode layer **50**A provided in this portion in the length direction L and the lamination direction T.
[0063] In cases where the second base electrode layer **50**B is also provided on a portion of at least one of the first main surface TS**1** or the second main surface TS**2**, the thickness of the second base electrode layer **50**B provided in this portion in the lamination direction T is, for example, preferably between about 3 μm and about 40 μm inclusive at the center of the second base electrode layer **50**B provided in this portion in the length direction L and the width direction W.
[0064] In cases where the second base electrode layer **50**B is also provided on a portion of at least one of the first lateral surface WS**1** or the second lateral surface WS**2**, the thickness of the second base electrode layer **50**B provided in this portion in the width direction W is, for example, preferably between about 3 μm and about 40 μm inclusive at the center of the second base electrode layer **50**B provided in this portion in the length direction L and the lamination direction T.
[0065] The first plated layer **60**A is provided to cover the first base electrode layer **50**A.
[0066] The second plated layer **60**B is provided to cover the second base electrode layer **50**B.
[0067] The first plated layer **60**A and the second plated layer **60**B may include at least one of, for example, Cu, Ni, Sn, Ag, Pd, Ag—Pd alloy, or Au. The first plated layer **60**A and the second plated layer **60**B may include a plurality of layers. The first plated layer **60**A and the second plated layer **60**B preferably have a two-portion structure including, for example, a Sn plated layer on top of a Ni plated layer.
[0068] The first plated layer **60**A is provided to cover the first base electrode layer **50**A. In the present example embodiment, the first plated layer **60**A includes, for example, a first Ni plated layer **61**A and a first Sn plated layer **62**A provided on the first Ni plated layer **61**A.
[0069] The second plated layer **60**B is provided to cover the second base electrode layer **50**B.
[0070] In the present example embodiment, the second plated layer **60**B includes, for example, a second Ni plated layer **61**B and a second Sn plated layer **62**B provided on the second Ni plated layer **61**B.
[0071] The Ni plated layer prevents the first base electrode layer **50**A and the second base electrode layer **50**B from being eroded by solder when mounting the multilayer ceramic capacitor **1**. The Sn plated layer improves the wettability of solder when mounting the multilayer ceramic capacitor **1**. As a result, the multilayer ceramic capacitor **1** can be easily mounted. The thickness of the first Ni plated layer **61**A, the first Sn plated layer **62**A, the second Ni plated layer **61**B, and the second Sn plated layer **62**B is, for example, preferably between about 1 μm and about 15 μm inclusive.
[0072] The external electrode **40** of example embodiment may include an electrically conductive

resin layer including electrically conductive particles and a thermosetting resin, for example. The electrically conductive resin layer may cover the fired layer. When the electrically conductive resin layer covers the fired layer, the electrically conductive resin layer is provided between the fired layer and the plated layers (the first plated layer **60**A and the second plated layer **60**B). The electrically conductive resin layer may completely cover the fired layer or may partially cover the fired layer.
[0073] The electrically conductive resin layer including a thermosetting resin is more flexible than an electrically conductive layer made of, for example, a plated film or a fired product of an electrically conductive paste. Therefore, even when an impact caused by physical shock or thermal cycle is applied to the multilayer ceramic capacitor **1**, the electrically conductive resin layer defines and functions as a buffer layer. Therefore, the electrically conductive resin layer reduces or prevents the occurrence of cracking in the multilayer ceramic capacitor **1**.
[0074] Metals of the electrically conductive particles may be, for example, Ag, Cu, Ni, Sn, Bi or alloys thereof. The electrically conductive particle preferably includes Ag, for example. The electrically conductive particle is a metal powder of Ag, for example. Ag is suitable as an electrode material because of its lowest resistivity among metals. In addition, since Ag is a noble metal, it is not likely to be oxidized, and weatherability thereof is high. Therefore, for example, the metal powder of Ag is suitable as the electrically conductive particle.
[0075] Furthermore, the electrically conductive particle may be, for example, a metal powder coated on the surface of the metal powder with Ag. When using those coated with Ag on the surface of the metal powder, the metal powder is preferably, for example, Cu, Ni, Sn, Bi, or an alloy powder thereof. In order to make the metal of the base material inexpensive while keeping the characteristics of Ag, it is preferable to use a metal powder coated with Ag, for example.
[0076] Furthermore, the electrically conductive particle may be formed by subjecting Cu and Ni to an oxidation prevention treatment. Furthermore, the electrically conductive particle may be, for example, a metal powder coated with Sn, Ni, and Cu on the surface of the metal powder. When using those coated with Sn, Ni, and Cu on the surface of the metal powder, the metal powder is preferably, for example, Ag, Cu, Ni, Sn, Bi, or an alloy powder thereof.
[0077] The shape of the electrically conductive particle is not particularly limited. For the electrically conductive particle, a spherical metal powder, a flat metal powder, or the like can be used. However, it is preferable to use a mixture of a spherical metal powder and a flat metal powder.
[0078] The electrically conductive particles included in the electrically conductive resin layer mainly ensure the conductivity of the electrically conductive resin layer. Specifically, by a plurality of electrically conductive particles being in contact with each other, an energization path is provided inside the electrically conductive resin layer.
[0079] The resin of the electrically conductive resin layer may
[0080] include, for example, at least one of a variety of known thermosetting resins such as epoxy resin, phenolic resin, urethane resin, silicone resin, polyimide resin, and the like. Among those, epoxy resin is excellent in heat resistance, moisture resistance, adhesion, etc., and thus is one of the more preferable resins. Furthermore, it is preferable that the resin of the electrically conductive resin layer includes a curing agent together with a thermosetting resin. When epoxy resin is used as a base resin, the curing agent for the epoxy resin may be various known compounds such as, for example, phenols, amines, acid anhydrides, imidazoles, active esters, and amide-imides.
[0081] The electrically conductive resin layer may include a plurality of layers. The thickest portion of the electrically conductive resin layer is preferably, for example, about 10 μm or more and about 150 μm or less.
[0082] The basic configuration of the multilayer ceramic capacitor **1** according to the present example embodiment has been described above. The dimension of the multilayer ceramic capacitor **1** including the multilayer body **10** and the external electrodes **40** in the length direction, denoted as

the L dimension, is preferably, for example, between about 0.2 mm and about 10 mm inclusive. The dimension of the multilayer ceramic capacitor **1** in the lamination direction, denoted as the T dimension, is preferably, for example, between about 0.05 mm and about 10 mm inclusive. The dimension of the multilayer ceramic capacitor **1** in the width direction, denoted as the W dimension, is preferably, for example, between about 0.1 mm and about 10 mm inclusive.

[0083] The multilayer ceramic capacitor **1** of the present example embodiment: including the basic configuration has the following characteristics in the external electrodes **40**, which is the first external electrode **40**A and the second external electrode **40**B.

[0084] The external electrodes **40** of the present example embodiment include the main surface-side external electrodes provided on at least one of the first main surface TS**1** or the second main surface TS**2**. Specifically, as described above, the first external electrode **40**A of the present example embodiment is provided on the first end surface LS**1** and extends from the first end surface LS**1** to a portion of the first main surface TS**1**, a portion of the second main surface TS**2**, a portion of the first lateral surface WS**1**, and a portion of the second lateral surface WS**2**. The first external electrode **40**A of the present example embodiment includes a first end surface-side external electrode **400**A as an end surface-side external electrode provided on the first end surface LS**1**, a first main surface-side external electrode **411**A as a main surface-side external electrode provided on the first main surface TS**1**, a second main surface-side external electrode **412**A as a main surface-side external electrode provided on the second main surface TS**2**, as illustrated in FIG. **2**, a first lateral surface-side external electrode **421**A provided on the first lateral surface WS**1**, and a second lateral surface-side external electrode **422**A provided on the second lateral surface WS**2**, as illustrated in FIGS. **4**A and **4**B.

[0085] As described above, the first external electrode **40**A includes the first base electrode layer **50**A and the first plated layer **60**A provided on the first base electrode layer **50**A. In the present example embodiment, the first base electrode layer **50**A extends from the first end surface LS**1** to a portion of the first main surface TS**1**, a portion of the second main surface TS**2**, a portion of the first lateral surface WS**1**, and a portion of the second lateral surface WS**2**, in which the first plated layer **60**A is provided to cover the first base electrode layer **50**A.

[0086] Specifically, as illustrated in FIG. **2**, the first end surface-side external electrode **400**A of the present example embodiment includes a first end surface-side base electrode layer **500**A provided on the first end surface LS**1**, and a first end surface-side plated layer **600**A provided above the first end surface-side base electrode layer **500**A. The first end surface-side base electrode layer **500**A is a portion of the first base electrode layer **50**A. The first end surface-side plated layer **600**A is a portion of the first plated layer **60**A, and includes the first Ni plated layer **61**A and the first Sn plated layer **62**A provided on the first Ni plated layer **61**A.

[0087] As illustrated in FIG. **2**, the first main surface-side external electrode **411**A of the present example embodiment includes a first main surface-side base electrode layer **511**A as a main surface-side base electrode layer provided on the first main surface TS**1**, and a first main surface-side plated layer **611**A as a main surface-side plated layer provided above the first main surface-side base electrode layer **511**A. The first main surface-side base electrode layer **511**A is a portion of the first base electrode layer **50**A. The first main surface-side plated layer **611**A is a portion of the first plated layer **60**A, and includes the first Ni plated layer **61**A and the first Sn plated layer **62**A provided on the first Ni plated layer **61**A.

[0088] As illustrated in FIG. **2**, the second main surface-side external electrode **412**A of the present example embodiment includes a second main surface-side base electrode layer **512**A as a main surface-side base electrode layer provided on the second main surface TS**2**, and a second main surface-side plated layer **612**A as a main surface-side plated layer provided above the second main surface-side base electrode layer **512**A. The second main surface-side base electrode layer **512**A is a portion of the first base electrode layer **50**A. The second main surface-side plated layer **612**A is a portion of the first plated layer **60**A, and includes the first Ni plated layer **61**A and the first Sn

plated layer **62**A provided on the first Ni plated layer **61**A.
[0089] As illustrated in FIGS. **4**A and **4**B, the first lateral surface-side external electrode **421**A of the present example embodiment includes a first lateral surface-side base electrode layer **521**A provided on the first lateral surface WS**1**, and a first lateral surface-side plated layer **621**A provided above the first lateral surface-side base electrode layer **521**A. The first lateral surface-side base electrode layer **521**A is a portion of the first base electrode layer **50**A. The first lateral surface-side plated layer **621**A is a portion of the first plated layer **60**A, and includes the first Ni plated layer **61**A and the first Sn plated layer **62**A provided on the first Ni plated layer **61**A.
[0090] As illustrated in FIGS. **4**A and **4**B, the second lateral surface-side external electrode **422**A of the present example embodiment includes a second lateral surface-side base electrode layer **522**A provided on the second lateral surface WS**2**, and a second lateral surface-side plated layer **622**A provided above the second lateral surface-side base electrode layer **522**A. The second lateral surface-side base electrode layer **522**A is a portion of the first base electrode layer **50**A. The second lateral surface-side plated layer **622**A is a portion of the first plated layer **60**A, and includes the first Ni plated layer **61**A and the first Sn plated layer **62**A provided on the first Ni plated layer **61**A.
[0091] The thickness of the first Ni plated layer **61**A and the first Sn plated layer **62**A of the first main surface-side plated layer **611**A, the second main surface-side plated layer **612**A, the first lateral surface-side plated layer **621**A, and the second lateral surface-side plated layer **622**A is preferably, for example, between about 1 μm and about 4 μm inclusive.
[0092] As described above, the second external electrode **40**B of the present example embodiment is provided on the second end surface LS**2** and extends from the second end surface LS**2** to a portion of the first main surface TS**1**, a portion of the second main surface TS**2**, a portion of the first lateral surface WS**1**, and a portion of the second lateral surface WS**2**. Specifically, the second external electrode **40**B of the present example embodiment includes a second end surface-side external electrode **400**B as an end surface-side external electrode provided on the second end surface LS**1**, a first main surface-side external electrode **411**B provided as a main surface-side external electrode on the first main surface TS**1**, a second main surface-side external electrode **412**B as a main surface-side external electrode provided on the second main surface TS**2**, as illustrated in FIG. **2**, a first lateral surface-side external electrode **421**B provided on the first lateral surface WS**1**, and a second lateral surface-side external electrode **422**B provided on the second lateral surface WS**2**, as illustrated in FIGS. **4**A and **4**B.
[0093] As described above, the second external electrode **40**B includes the second base electrode layer **50**B and the second plated layer **60**B provided on the second base electrode layer **50**B. In the present example embodiment, the second base electrode layer **50**B extends from the second end surface LS**2** to a portion of the first main surface TS**1**, a portion of the second main surface TS**2**, a portion of the first lateral surface WS**1**, and a portion of the second lateral surface WS**2**, in which the second plated layer **60**B is provided to cover the second base electrode layer **50**B.
[0094] Specifically, as illustrated in FIG. **2**, the second end surface-side external electrode **400**B of the present example embodiment includes a second end surface-side base electrode layer **500**B provided on the second end surface LS**2**, and a second end surface-side plated layer **600**B provided above the second end surface-side base electrode layer **500**B. The second end surface-side base electrode layer **500**B is a portion of the second base electrode layer **50**B. The second end surface-side plated layer **600**B is a portion of the second plated layer **60**B, and includes the second Ni plated layer **61**B and the second Sn plated layer **62**B provided on the second Ni plated layer **61**B.
[0095] As illustrated in FIG. **2**, the first main surface-side
[0096] external electrode **411**B of the present example embodiment includes the first main surface-side base electrode layer **511**B as a main surface-side base electrode layer provided on the first main surface TS**1**, and the first main surface-side plated layer **611**B as a main surface-side plated layer provided above the first main surface-side base electrode layer **511**B. The first main surface-side base electrode layer **511**B is a portion of the second base electrode layer **50**B. The first main

surface-side plated layer **611**B is a portion of the second plated layer **60**B, and includes the second Ni plated layer **61**B and the second Sn plated layer **62**B provided on the second Ni plated layer **61**B.

[0097] As illustrated in FIG. **2**, the second main surface-side external electrode **412**B of the present example embodiment includes the second main surface-side base electrode layer **512**B as a main surface-side base electrode layer provided on the second main surface TS**2**, and the second main surface-side plated layer **612**B as a main surface-side plated layer provided above the second main surface-side base electrode layer **512**B. The second main surface-side base electrode layer **512**B is a portion of the second base electrode layer **50**B. The second main surface-side plated layer **612**B is a portion of the second plated layer **60**B, and includes the second Ni plated layer **61**B and the second Sn plated layer **62**B provided on the second Ni plated layer **61**B.

[0098] As illustrated in FIGS. **4**A and **4**B, the first lateral surface-side external electrode **421**B of the present example embodiment includes the first lateral surface-side base electrode layer **521**B provided on the first lateral surface WS**1** and the first lateral surface-side plated layer **621**B provided above the first lateral surface-side base electrode layer **521**B. The first lateral surface-side base electrode layer **621**B is a portion of the second base electrode layer **50**B. The first lateral surface-side plated layer **621**B is a portion of the second plated layer **60**B, and includes the second Ni plated layer **61**B and the second Sn plated layer **62**B provided on the second Ni plated layer **61**B.

[0099] As illustrated in FIGS. **4**A and **4**B, the second lateral surface-side external electrode **422**B of the present example embodiment includes the second lateral surface-side base electrode layer **522**B provided on the second lateral surface WS**2** and the second lateral surface-side plated layer **622**B provided above the second lateral surface-side base electrode layer **522**B. The second lateral surface-side base electrode layer **522**B is a portion of the second base electrode layer **50**B. The second lateral surface-side plated layer **622**B is a portion of the second plated layer **60**B, and includes the second Ni plated layer **61**B and the second Sn plated layer **62**B provided on the second Ni plated layer **61**B.

[0100] The thickness of the second Ni plated layer **61**B and the second Sn plated layer **62**B of the first main surface-side plated layer **611**B, the second main surface-side plated layer **612**B, the first lateral surface-side plated layer **621**B, and the second lateral surface-side plated layer **622**B is preferably, for example, between about 1 μm and about 4 μm inclusive.

[0101] FIG. **2** illustrates the LT cross-sectional view of the multilayer ceramic capacitor **1** and the multilayer body **10** along the lamination direction T and the length direction L. In the LT cross-sectional view, the first main surface-side external electrode **411**A of the first external electrode **40**A includes a first main surface-side recess **510**A as a recess recessed towards the multilayer body **10** side. The first main surface-side recess **510**A is provided on the surface of the first main surface-side external electrode **411**A. The shape of the first main surface-side recess **510**A is a groove extending in the width direction W orthogonal or substantially orthogonal to the LT cross-section, i.e., in the direction perpendicular to the paper surface of FIG. **2**. The first main surface-side recess **510**A may be provided over the entire or substantially the entire length of the first main surface-side external electrode **411**A along the width direction W. The first main surface-side recess **510**A is provided in or adjacent to the approximate center of the first main surface-side external electrode **411**A in the length direction L.

[0102] As illustrated in FIG. **2**, the second main surface-side external electrode **412**A of the first external electrode **40**A includes a second main surface-side recess **520**A as a recess recessed towards the multilayer body **10** side in the LT cross-sectional view. The second main surface-side recess **520**A is provided on the surface of the second main surface-side external electrode **412**A. The shape of the second main surface-side recess **520**A is a groove extending in the width direction W orthogonal or substantially orthogonal to the LT cross-section, i.e., in the direction perpendicular or substantially perpendicular to the paper surface of FIG. **2**. The second main surface-side recess

520A may be provided over the entire or substantially the entire length of the second main surface-side external electrode **412**A along the width direction W. The second main surface-side recess **520**A is provided in or adjacent to the approximate center of the second main surface-side external electrode **412**A in the length direction L.

[0103] As illustrated in FIG. **2**, the first main surface-side external **411**B of the second external electrode **40**B includes a first main surface-side recess **510**B as a recess recessed towards the multilayer body **10** side in the LT cross-sectional view. The first main surface-side recess **510**B is provided on the surface of the first main surface-side external electrode **411**B. The shape of the first main surface-side recess **510**B is a groove extending in the width direction W orthogonal or substantially orthogonal to the LT cross-section, i.e., in the direction perpendicular or substantially perpendicular to the paper surface of FIG. **2**. The first main surface-side recess **510**B may be provided over the entire or substantially the entire length of the first main surface-side external electrode **411**B along the width direction W. The first main surface-side recess **510**B is provided in or adjacent to the approximate center of the first main surface-side external electrode **411**B in the length direction L.

[0104] As illustrated in FIG. **2**, the second main surface-side external electrode **412**B of the second external electrode **40**B includes a second main surface-side recess **520**B as a recess recessed towards the multilayer body **10** side in the LT cross-sectional view. The second main surface-side recess **520**B is provided on the surface of the second main surface-side external electrode **412**B. The shape of the second main surface-side recess **520**B is a groove extending in the width direction W orthogonal or substantially orthogonal to the LT cross-section, i.e., in the direction perpendicular or substantially perpendicular to the paper surface of FIG. **2**. The second main surface-side recess **520**B may be provided over the entire or substantially the entire length of the second main surface-side external electrode **412**B along the width direction W. The second main surface-side recess **520**B is provided in or adjacent to the approximate center of the second main surface-side external electrode **412**B in the length direction L.

[0105] FIGS. **4**A and **4**B illustrate LW cross-sectional views of the multilayer ceramic capacitor **1** and the multilayer body **10** along the length direction L and the width direction W. In the LW cross-sectional views, the first lateral surface-side external electrode **421**A of the first external electrode **40**A includes a first lateral surface-side recess **530**A as a recess recessed towards the multilayer body **10** side. The first lateral surface-side recess **530**A is provided on the surface of the first lateral surface-side external electrode **421**A. The shape of the first lateral surface-side recess **530**A is a groove extending in the lamination direction T orthogonal or substantially orthogonal to the LW cross-section, i. e., in the direction perpendicular or substantially perpendicular to the paper surface of FIGS. **4**A and **4**B. The first lateral surface-side recess **530**A may be provided over the entire or substantially the entire length of the first lateral surface-side external electrode **421**A along the lamination direction T. The first lateral surface-side recess **530**A is provided in or adjacent to the approximate center of the first lateral surface-side external electrode **421**A in the length direction L. The first lateral surface-side recess **530**A may communicate with either or both of the first main surface-side recess **510**A and the second main surface-side recess **520**A, or may not communicate with both.

[0106] As illustrated in FIGS. **4**A and **4**B, the second lateral surface-side external electrode **422**A of the first external electrode **40**A includes a second lateral surface-side recess **540**A as a recess recessed towards the multilayer body **10** side in the LW cross-sectional view. The second lateral surface-side recess **540**A is provided on the surface of the second lateral surface-side external electrode **422**A. The shape of the second lateral surface-side recess **540**A is a groove extending in the lamination direction T orthogonal or substantially orthogonal to the LW cross-section, i.e., in the direction perpendicular or substantially perpendicular to the paper surface of FIGS. **4**A and **4**B. The second lateral surface-side recess **540**A may be provided over the entire or substantially the entire length of the second lateral surface-side external electrode **422**A along the lamination direction T. The second lateral surface-side recess **540**A is provided in or adjacent to the approximate center of the second lateral surface-side external electrode **422**A in the length direction L. The second lateral surface-side recess **540**A may communicate with either or both of the first main surface-side recess **510**A and the second main surface-side recess **520**A, or may not communicate with both.

[0107] As illustrated in FIGS. **4**A and **4**B, the first lateral surface-side external electrode **421**B of the second external electrode **40**B includes a first lateral surface-side recess **530**B as a recess recessed towards the multilayer body **10** side in the LW cross-sectional view. The first lateral surface-side recess **530**B is provided on the surface of the first lateral surface-side external electrode **421**B. The shape of the first lateral surface-side recess **530**B is a groove extending in the lamination direction T orthogonal or substantially orthogonal to the LW cross-section, i.e., in the direction perpendicular or substantially perpendicular to the paper surface of FIGS. **4**A and **4**B. The first lateral surface-side recess **530**B may be provided over the entire or substantially the entire length of the first lateral surface-side external electrode **421**B along the lamination direction T. The first lateral surface-side recess **530**B is provided in or adjacent to the approximate center of the first lateral surface-side external electrode **421**B in the length direction L. The first lateral surface-side recess **530**B may communicate with either or both of the first main surface-side recess **510**B and the second main surface-side recess **520**B, or may not communicate with both.

[0108] As illustrated in FIGS. **4**A and **4**B, the second lateral surface-side external electrode **422**B of the second external electrode **40**B includes a second lateral surface-side recess **540**B as a recess recessed towards the multilayer body **10** side in the LW cross-sectional view. The second lateral surface-side recess **540**B is provided on the surface of the second lateral surface-side external electrode **422**B. The shape of the second lateral surface-side recess **540**B is a groove extending in the lamination direction T orthogonal or substantially orthogonal to the LW cross-section, i.e., in the direction perpendicular or substantially perpendicular to the paper surface of FIGS. **4**A and **4**B. The second lateral surface-side recess **540**B may be provided over the entire or substantially the entire length of the second lateral surface-side external electrode **422**B along the lamination direction T. The second lateral surface-side recess **540**B is provided in or adjacent to the approximate center of the second lateral surface-side external electrode **422**B in the length direction L. The second lateral surface-side recess **540**B may communicate with either or both of the first main surface-side recess **510**AB and the second main surface-side recess **520**AB, or may not communicate with both.

[0109] The first main surface-side external electrode **411**A and the second main surface-side external electrode **412**A of the first external electrode **40**A, as well as the first main surface-side external electrode **411**B and the second main surface-side external electrode **412**B of the second external electrode **40**B, share the same or substantially the same configuration. Similarly, the first lateral surface-side external electrode **421**A and the second lateral surface-side external electrode **422**A of the first external electrode **40**A, as well as the first lateral surface-side external electrode **421**B and the second lateral surface-side external electrode **422**B of the second external electrode **40**B, also share the same or substantially the same configuration as the four main surface-side external electrodes **411**A, **412**A, **411**B, and **412**B.

[0110] The first main surface-side recess **510**A and the second main surface-side recess **520**A of the first external electrode **40**A, as well as the first main surface-side recesses **510**B and the second main surface-side recesses **520**B of the second external electrode **40**B, share the same or substantially the same configuration. Similarly, the first lateral surface-side recess **530**A and the second lateral surface-side recess **540**A of the first external electrode **40**A, as well as the first lateral surface-side recess **530**B and the second lateral surface-side recess **540**B of the second external electrode **40**B, also share the same or substantially the same configuration as the four main surface-side recesses **510**A, **520**A, **510**B, and **520**B.

[0111] Accordingly, the first main surface-side external electrode **411**A and the first main surface-side recess **510**A of the first external electrode **40**A will be described below, representatively describing the four main surface-side external electrodes, the four main surface-side recesses, the four lateral surface-side external electrodes, and the four lateral surface-side recesses.

[0112] The first main surface-side external electrode **411**A of the first external electrode **40**A corresponds to the second main surface-side external electrode **412**A, the first lateral surface-side external electrode **421**A, and the second lateral surface-side external electrode **422**A of the first external electrode **40**A, as well as the first main surface-side external electrode **411**B, the second main surface-side external electrode **412**B, the first lateral surface-side external electrode **421**B, and the second lateral surface-side external electrode **422**B of the second external electrode **40**B. The first main surface-side recess **510**A of the first external electrode **40**A corresponds to the second main surface-side recess **520**A, the first lateral surface-side recess **530**A, and the second lateral surface-side recess **540**A of the first external electrode **40**A, as well as the first main surface-side recesses **510**B, the second main surface-side recesses **520**B, the first lateral surface-side recess **530**B, and the second lateral surface-side recess **540**B of the second external electrode **40**B.

[0113] The first base electrode layer **50**A and the first plated layer **60**A of the first external electrode **40**A correspond to the second base electrode layer **50**B and the second plated layer **60**B of the second external electrode **40**B. The first Ni plated layer **61**A and the first Sn plated layer **62**A of the first plated layer **60**A of the first external electrode **40**A correspond to the second Ni plated layer **61**B and the second Sn plated layer **62**B of the second plated layer **60**B of the second external electrode **40**B.

[0114] FIG. **5**A is an enlarged view of a portion indicated by VA in FIG. **2**, illustrating an LT cross-sectional view of the first main surface-side external electrode **411**A of the first external electrode **40**A. FIG. **5**B corresponds to FIG. **5**A and illustrates the outer shape of the LT cross-sectional view of the first main surface-side external electrode **411**A of the first external electrode **40**A. FIGS. **5**A and **5**B illustrate the same XYZ orthogonal coordinate system as illustrated in FIGS. **1** to **4**B.

[0115] As illustrated in FIG. **5**A, the first main surface-side external electrode **411**A of the first external electrode **40**A includes the first main surface-side base electrode layer **511**A provided on the first main surface TS**1**, and the first main surface-side plated layer **611**A including the first Ni plated layer **61**A and the first Sn plated layer **62**A. The three layers, which include the first Sn plated layer **62**A of the outermost first main surface-side plated layer **611**A, the first Ni plated layer **61**A as a lower layer below the first Sn plated layer **62**A, and the first main surface-side base electrode layer **511**A as a lower layer below the first Ni plated layer **61**A, are recessed towards the multilayer body **10** side in the lamination direction T (the Z direction in FIGS. **5**A and **5**B), thus defining the first main surface-side recess **510**A.

[0116] Therefore, the thickness of the first main surface-side base electrode layer **511**A in the lamination direction T is minimized in the portion corresponding to the first main surface-side recess **510**A. In the present example embodiment, the maximum thickness of the first main surface-side base electrode layer **511**A is preferably, for example, between about 15 μm and about 30 μm inclusive.

[0117] As illustrated in FIG. **5**A, in the LT cross-sectional view, the main surface-side external electrode **411**A of the first external electrode **40**A includes the first main surface-side recess **510**A, an inner peripheral region **560**, and an outer peripheral region **570** as peripheral regions adjacent to the first main surface-side recess **510**A in the length direction L (corresponding to the X direction in FIG. **5**A).

[0118] The first main surface-side recess **510**A is a portion of the region **700** recessed towards the multilayer body **10** side in the length direction L, as will be described later with reference to FIG. **5**B, and is illustrated as a recessed region **550** in FIG. **5**A.

[0119] The inner peripheral region **560** within the first main surface-side external electrode **411**A is a region to the inner side of the first main surface-side recess **510**A in the length direction L, specifically a region to the central side of the multilayer body **10** in the length direction L (to the side distant from the first end surface LS**1** in the length direction L), continuing from the recessed region **550** over a range substantially equivalent to the recessed region **550** in the length direction L.

[0120] The outer peripheral region **570** within the first main surface-side external electrode **411**A is a region to the outer side of the first main surface-side recess **510**A in the length direction L, specifically a region to the outer side of the multilayer body **10** in the length direction L (to the side closer to the first end surface LS**1** in the length direction L), continuing from the recessed region **550** over a range equivalent or substantially equivalent to the recessed region **550** in the length direction L.

[0121] As illustrated in FIG. **5**B, the surface of the first main surface-side external electrode **411**A includes the first main surface-side recess **510**A, an inner bulge **710** extending to the inner side of the first main surface-side recess **510**A, and a second bulge **720** extending to the outer side of the first main surface-side recess **510**A. The first bulge **710** is a region corresponding to the inner peripheral region **560**. The second bulge **720** is a region corresponding to the outer peripheral region **570**.

[0122] The reference number **700** in FIG. **5**B represents a region of the first main surface-side recess **510**A (recessed region **550**) in the length direction L, in the present example embodiment. The region **700** of the first main surface-side recess **510**A in the length direction L is based on the distance in the length direction L between a first midpoint **510***m***1**, which connects a deepest portion **510***d* of the first main surface-side recess **510**A to a vertex **710***p* of the first bulge **710**, and a second midpoint **510***m***2**, which connects the deepest portion **510***d* of the first main surface-side recess **510**A to a vertex **720***p* of the second bulge **720**. The deepest portion **510***d* of the first main surface-side recess **510**A refers to the portion closest to the first main surface TS**1** of the multilayer body **10** in the lamination direction T, in the first main surface-side recess **510**A. The vertex **710***p* of the first bulge **710** is the point farthest from the first main surface TS**1** of the multilayer body **10** in the lamination direction T, on the surface of the first bulge **710**. The vertex **720***p* of the second bulge **720** is the point farthest from the first main surface TS**1** of the multilayer body **10** in the lamination direction T, on the surface of the second bulge **720**.

[0123] The distance between the vertex **710***p* of the first bulge **710** and the first main surface TS**1** of the multilayer body **10** in the lamination direction T is the height **710**H of the first bulge **710**. The distance between the vertex **720***p* of the second bulge **720** and the first main surface TS**1** of the multilayer body **10** in the lamination direction T is the height **720**H of the second bulge **720**. In the present example embodiment, the height **710**H of the first bulge **710** and the height **720**H of the second bulge **720** may be the same or different. In the cases of being different, the height **710**H of the first bulge **710** may be higher or lower than the height **720**H of the second bulge **720**.

[0124] As illustrated in FIG. **5**B, in the present example embodiment, the depth D of the first main surface-side recess **510**A refers to the shortest distance between the deepest portion **510***d* and the line connecting the vertex **710***p* of the first bulge **710** and the vertex **720***p* of the second bulge **720**.

[0125] In the present example embodiment, the depth D of the first main surface-side recess **510**A is preferably, for example, between about 3 μm and about 10 μm inclusive.

[0126] In the present example embodiment, the depth D of the first main surface-side recess **510**A is preferably greater than the thickness of the Ni plated layer **61**A in the lamination direction T.

[0127] As illustrated in FIG. **5**B, in the present example embodiment, the distance **730**L between the vertex **710***p* of the first bulge **710** and the vertex **720***p* of the second bulge **720** in the length direction L is preferably, for example, between about 120 μm and about 150 μm inclusive. The distance **730**L is preferably greater than the maximum thickness of the first main surface-side base electrode layer **511**A.

[0128] As illustrated in FIG. **5**B, in the present example embodiment, the distance **740**L between the inner end **560***a* of the first main surface-side base electrode layer **511**A and the deepest portion **510***d* of the first main surface-side recess **510**A in the length direction L is preferably, for example, between about 110 μm and about 130 μm inclusive.

[0129] As illustrated in FIG. 5B, in the present example embodiment, the distance **750**L between the inner end **560***a* of the first main surface-side base electrode layer **511**A and the vertex **710***p* of the first bulge **710** in the length direction L is preferably, for example, between about 70 μm and about 90 μm inclusive.

[0130] In the first main surface-side external electrode **411**A of the present example embodiment, the surface roughness Ra of the surface **550***s* of the first main surface-side recess **510**A is rougher than the surface roughness Ra of the surface **560***s* of the inner peripheral region **560** corresponding to the first bulge **710**, and rougher than the surface roughness Ra of the surface **570***s* of the outer peripheral region **570** corresponding to the second bulge **720**.

[0131] The surface roughness Ra of the surface **550***s* of the first main surface-side recess **510**A is preferably, for example, about 0.7 μm or more, and more preferably between about 0.7 μm and about 1.4 μm inclusive. The surface roughness Ra is even more preferably, for example, between about 1.0 μm and about 1.4 μm inclusive.

[0132] The surface roughness Ra of the surface **560***s* of the inner peripheral region **560** and the surface roughness Ra of the surface **570***s* of the outer peripheral region **570** are preferably, for example, about 0.5 μm or less, and more preferably between about 0.2 μm and about 0.5 μm inclusive. The surface roughness Ra is even more preferably, for example, between about 0.4 μm and about 0.5 μm inclusive.

[0133] In the multilayer ceramic capacitor **1** of the present example embodiment, the first main surface-side external electrode **411**A of the first external electrode **40**A includes the first main surface-side recess **510**A. The first main surface-side external electrode **411**A includes the inner peripheral region **560** as an inner region in the length direction L, and the outer peripheral region **570** as an outer region in the length direction L, in relation to the first main surface-side recess **510**A. The surface roughness Ra of the surface **550***s* of the first main surface-side recess **510**A is rougher than the surface roughness Ra of the surface **560***s* of the inner peripheral region **560**, and rougher than the surface roughness Ra of the surface **570***s* of the outer peripheral region **570**.

[0134] The multilayer ceramic capacitor **1** of the present example embodiment is mounted on a board. Mounting on the board may involve, for example, soldering the external electrodes **40** to the terminals of the board. In the case of attaching the first main surface-side external electrode **411**A to the board by soldering, the flexural stress occurring in the first main surface-side external electrode **411**A particularly concentrates on the inner end portion **411***e* of the first main surface-side external electrode **411**A illustrated in FIG. 5B, or on the inner end **560***a* of the first main surface-side base electrode layer **511**A, and is transmitted as tensile stress to the multilayer body **10**; as a result, cracks or the like may occur in the multilayer body **10**. In the multilayer ceramic capacitor **1** of the present example embodiment, since the first main surface-side external electrode **411**A includes the first main surface-side recess **510**A, a recess corresponding to the first main surface-side recess **510**A is also provided in the first main surface-side base electrode layer **511**A. Consequently, the total amount of the first main surface-side base electrode layer **511**A can be reduced compared to the case of lacking the first main surface-side recess **510**A. The reduction in the total amount results in a decrease in the flexural stress caused by the tensile stress concentrated at the inner end portion **411***e* or the end **560***a*. This enables improved flexural resistance of the multilayer ceramic capacitor **1**, and consequently reduces or prevents the occurrence of cracks or the like in the multilayer body **10**.

[0135] When the first main surface-side external electrode **411**A is soldered to the board, solder enters between the first main surface-side external electrode **411**A and the board. This solder comes into contact with the surface of the first main surface-side external electrode **411**A, and a larger contact area causes greater bending stress to the first main surface-side external electrode **411**A, leading to a risk of cracks or the like occurring in the multilayer body **10**. In the present example embodiment, the surface roughness Ra of the surface **550***s* of the first main surface-side recess

510A is rougher than the surface roughness Ra of the surface **560***s* of the inner peripheral region **560**, and rougher than the surface roughness Ra of the surface **570***s* of the outer peripheral region **570**. As a result, the solder is less likely to wet and fully contact the surface **550***s* of the first main surface-side recess **510**A, and the contact area of the solder with the surface **550***s* of the first main surface-side recess **510**A is smaller than the surface **560***s* of the inner peripheral region **560** and the surface **570***s* of the outer peripheral region **570**. Consequently, reducing the contact area of the solder with the entire or substantially the entire surface of the first main surface-side external electrode **411**A enables a reduction in the tensile stress caused by the bending stress concentrated at the inner end portion **411***e* or the end **560***a*, and consequently reduces or prevents the occurrence of cracks or the like in the multilayer body **10**. The bonding force of the solder is secured by the surface **560***s* of the inner peripheral region **560** and the surface **570***s* of the outer peripheral region **570**, where the contact area of the solder is relatively large.
[0136] Next, an example of a method of measuring the surface roughness Ra of the first main surface-side external electrode **411**A, and the depth D of the first main surface-side recess **510**A will be described.
[0137] First, the method of measuring the surface roughness Ra (arithmetic mean roughness Ra) of the first main surface-side external electrode **411**A will be described. Using a laser microscope (for example, manufactured by Keyence Corporation (registered trademark), VK-X1000, magnification 20×), the surfaces of the first main surface-side recess **510**A (recessed region **550**), the inner peripheral region **560**, and the outer peripheral region **570** are scanned with a laser to obtain images of each surface. Subsequently, the surface roughness Ra of each surface is measured using analysis software (for example, manufactured by Keyence Corporation (registered trademark), Multi-File Analysis Application).
[0138] Next, the following describes an example of a method of measuring the depth D of the first main surface-side recess **510**A.
[0139] The multilayer ceramic capacitor **1** is polished from either the first lateral surface WS**1** or the second lateral surface WS**2** to approximately half the width dimension W. As a result, the LT cross section is exposed in the middle of the multilayer ceramic capacitor **1** in the width direction W. Next, the depth D of the first main surface-side recess **510**A in the LT cross section exposed by polishing is measured using a digital microscope. As a result, the depth D of the first main surface-side recess **510**A is confirmed.
[0140] Next, an example of a method of manufacturing the multilayer ceramic capacitor **1** according to the present example embodiment will be described. The method of manufacturing the multilayer ceramic capacitor **1** according to the present example embodiment is not limited, as long as the requirements described above are satisfied. However, a preferable manufacturing method includes the following steps. The details of each step are described below.
[0141] A dielectric sheet for the dielectric layer **20** and a conductive paste for the internal electrode layer **30** are prepared. Both of the dielectric sheet for the dielectric layer **20** and the conductive paste for the internal electrode layer **30** include binders and solvents. The binders and solvents may be any known ones. The conductive paste is, for example, obtained by adding organic binders and organic solvents to metal powders.
[0142] The conductive paste for the internal electrode layer **30** is printed in a predetermined pattern on the dielectric sheet, for example, by screen printing or gravure printing. As a result, a dielectric sheet with a pattern of the first internal electrode layer **31** and a dielectric sheet with a pattern of the second internal electrode layer **32** are prepared.
[0143] A predetermined number of dielectric sheets without a printed pattern of the internal electrode layer **30** are stacked, thus forming a portion that becomes the first main surface-side outer layer portion **12** on the first main surface TS**1** side. On top of this, the dielectric sheet with the pattern of the first internal electrode layer **31** and the dielectric sheet with the pattern of the second internal electrode layer **32** are sequentially stacked alternately. As a result, a portion that becomes

the inner layer portion **11** is formed. A predetermined number of dielectric sheets without the printed pattern of the internal electrode layer **30** are stacked on top of the portion that becomes the inner layer portion **11**, thereby forming a portion that becomes the second main surface-side outer layer portion **13** on the second main surface TS**2** side. As a result, a multilayer sheet is produced.
[0144] The multilayer sheet is pressed in the lamination direction, for example, by hydrostatic pressure pressing, to produce a multilayer block.
[0145] The multilayer block is cut into individual pieces to obtain a plurality of multilayer chips. Subsequently, the multilayer chips may be polished, for example, by barrel polishing, to round the corners and edges.
[0146] The multilayer chips are fired to produce the multilayer body **10**. The firing temperature in this case is preferably, for example, between about 900° C. and about 1400° C. inclusive, depending on the materials of the dielectric layer **20** and the internal electrode layer **30**.
[0147] The first external electrode **40**A and the second external electrode **40**B are formed as follows on the end surfaces of the multilayer body **10**.
[0148] A conductive paste, which becomes the first base electrode layer **50**A, is applied to the first end surface LS**1** side of the multilayer body **10**. A conductive paste, which becomes the second base electrode layer **50**B, is applied to the second end surface LS**2** side of the multilayer body **10**. In the present example embodiment, the first base electrode layer **50**A and the second base electrode layer **50**B are fired layers. The fired layers are formed by applying a conductive paste containing glass components and metal to the multilayer body **10**, for example, by a method such as dipping, followed by firing. The firing temperature in this case is preferably, for example, between about 700° C. and about 950° C. inclusive.
[0149] The first base electrode layer **50**A and the second base electrode layer **50**B are preferably fired layers. This allows for relatively simply forming the first base electrode layer **50**A and the second base electrode layer **50**B, as compared to thin film formation methods such as, for example, sputtering or evaporation.
[0150] In the present example embodiment, a conductive paste, which becomes the first main surface-side base electrode layer **511**A of the first base electrode layer **50**A, is applied by dipping to extend from the first end surface LS**1** of the multilayer body **10** to a portion of the first main surface TS**1**, and a conductive paste, which becomes the second main surface-side base electrode layer **512**A of the first base electrode layer **50**A, is applied by dipping to extend from the first end surface LS**1** of the multilayer body **10** to a portion of the second main surface TS**2**.
[0151] Similarly, in the present example embodiment, a conductive paste, which becomes the first main surface-side base electrode layer **511**B of the second base electrode layer **50**B, is applied by dipping to extend from the second end surface LS**2** of the multilayer body **10** to a portion of the first main surface TS**1**, and a conductive paste, which becomes the second main surface-side base electrode layer **512**B of the second base electrode layer **50**B, is applied by dipping to extend from the second end surface LS**2** of the multilayer body **10** to a portion of the second main surface TS**2**.
[0152] In this case, a conductive paste, which becomes the first lateral surface-side base electrode layer **521**A of the first base electrode layer **50**A, is preferably applied by dipping to extend from the first end surface LS**1** of the multilayer body **10** to a portion of the first lateral surface WS**1**, and a conductive paste, which becomes the second lateral surface-side base electrode layer **522**A of the first base electrode layer **50**A, is preferably applied by dipping to extend from the first end surface LS**1** of the multilayer body **10** to a portion of the second lateral surface WS**2**.
[0153] Similarly, in this case, a conductive paste, which becomes the first lateral surface-side base electrode layer **521**B of the second base electrode layer **50**B, is preferably applied by dipping to extend from the second end surface LS**2** of the multilayer body **10** to a portion of the first lateral surface WS**1**, and a conductive paste, which becomes the second lateral surface-side base electrode layer **522**B of the second base electrode layer **50**B, is preferably applied by dipping to extend from the second end surface LS**2** of the multilayer body **10** to a portion of the second lateral surface

WS**2**.
[0154] The pre-firing multilayer chips and the conductive paste applied to the multilayer chips may be simultaneously fired. In this case, the fired layers are preferably formed by firing a material including a ceramic material instead of glass components. In this case, as the ceramic material to be added, a ceramic material of the same type as the dielectric layer **20** is preferably used. In this case, a conductive paste is applied to the pre-firing multilayer chips, and the multilayer chips as well as the conductive paste applied to the multilayer chips are simultaneously fired, thus forming the multilayer body **10** with the fired layers. The multilayer body **10** is fired simultaneously with forming the fired layers, allowing for simplifying the manufacturing steps.
[0155] Subsequently, a plating treatment is performed on the surfaces of the first base electrode layer **50**A and the second base electrode layer **50**B including the fired layers. In the present example embodiment, the first plated layer **60**A is formed on the surface of the first base electrode layer **50**A. The second plated layer **60**B is formed on the surface of the second base electrode layer **50**B. In the present example embodiment, the first Ni plated layer **61**A is formed on the surface of the first base electrode layer **50**A, and the first Sn plated layer **62**A is formed on the surface of the first Ni plated layer **61**A. In the present example embodiment, the second Ni plated layer **61**B is formed on the surface of the second base electrode layer **50**B, and the second Sn plated layer **62**B is formed on the surface of the second Ni plated layer **61**B.
[0156] In performing the plating treatment, either electrolytic plating or electroless plating may be used. However, electroless plating requires pretreatment with catalysts to improve the plating deposition rate, involving a drawback of increased complexity of the steps. Therefore, electrolytic plating is preferable in most cases. The Ni plated layer and the Sn plated layer are preferably sequentially formed, for example, by barrel plating.
[0157] When the electrically conductive resin layer is provided, the electrically conductive resin layer may cover the fired layer. When the electrically conductive resin layer is provided, an electrically conductive resin paste including, for example, a thermosetting resin and a metal component is applied on the fired layer, and then heat treatment is performed at a temperature of, for example, about 250° C. to about 550° C. or higher. Thus, the thermosetting resin is thermally cured to form the electrically conductive resin layer. The atmosphere during the heat treatment is preferably, for example, an N2 atmosphere. Furthermore, in order to prevent scattering of the resin and to prevent oxidation of various metal components, the oxygen concentration preferably, for example, about 100 ppm or less.
[0158] Here, for example, the following examples of methods may be used to obtain the recess with a groove shape on the main surface-side external electrode and the lateral surface-side external electrode of the external electrode **40**, as in the present example embodiment.
[0159] FIGS. **6**A to **6**C schematically illustrate the steps of forming the base electrode layer. Firstly, as illustrated in FIG. **6**A, a first conductive paste P**1**, which becomes the base electrode layer, is applied by dipping to the end of the multilayer body **10** in the length direction L. The first conductive paste P**1** used herein typically has relatively high viscosity. Next, as illustrated in FIG. **6**B, the first conductive paste P**1** at and near the end surface of the multilayer body **10** is removed, forming the first bulge **710**. Subsequently, as illustrated in FIG. **6**C, a second conductive paste P**2** with relatively low viscosity is shallowly applied by dipping up to the front of the first bulge **710**. As a result, the second bulge **720** is formed, and a recess G is formed between the second bulge **720** and the first bulge **710**. The recess G becomes the first main surface-side recess **510**A, the second main surface-side recess **520**A, the first lateral surface-side recess **530**A, the second lateral surface-side recess **540**A, the first main surface-side recess **510**B, the second main surface-side recess **520**B, the first lateral surface-side recess **530**B, and the second lateral surface-side recess **540**B.
[0160] For example, the following methods can be used to control the surface roughness of the main surface-side external electrode and the lateral surface-side external electrode of the external

electrodes **40** as in the present example embodiment, i.e., making the surface roughness of the recess rougher than the surface roughness of the peripheral regions.

[0161] Barrel polishing is performed after forming the external electrodes. The size of the recess, the size of the media used for barrel polishing, and the conditions of barrel polishing are adjusted, thus adjusting the contact conditions of the media with the surface of the recess and the surface the peripheral regions, ensuring that the surface of the recess becomes rougher than the surface of the peripheral regions. For example, barrel polishing is performed while masking the surface of the recess such that the surface of the recess is not polished and the surface of the peripheral regions is polished, such that the surface roughness of the recess can become rougher than the surface roughness of the peripheral regions.

[0162] The multilayer ceramic capacitor **1** may be manufactured through the manufacturing steps described above.

[0163] The configurations of the multilayer ceramic capacitor

[0164] **1** is not limited to those illustrated in FIGS. **1** to **4**B. For example, the multilayer ceramic capacitor **1** may include a two-portion structure, a three-portion structure, or a four-portion structure as illustrated in FIGS. **7**A to **7**C.

[0165] The multilayer ceramic capacitor **1** illustrated in FIG. **7**A includes a two-portion structure, equipped with a floating internal electrode layer **35** as a floating inner conductive layer that does not extend to either the first end surface LS**1** or the second end surface LS**2**, in addition to the first internal electrode layer **33** and the second internal electrode layer **34** as the internal electrode layer **30**.

[0166] The multilayer ceramic capacitor **1** illustrated in FIG. **7**B includes a three-portion structure, equipped with a first floating internal electrode layer **35**A and a second floating internal electrode layer **35**B, as the floating internal electrode layer **35**.

[0167] The multilayer ceramic capacitor **1** illustrated in FIG. **7**C includes a four-portion structure, equipped with the first floating internal electrode layer **35**A, the second floating internal electrode layer **35**B, and a third floating internal electrode layer **35**C, as the floating internal electrode layer **35**.

[0168] The multilayer ceramic capacitor **1** can be structured with a plurality of divided counter electrode portions by providing the floating internal electrode layers **35** as the internal electrode layer **30**. As a result, a plurality of capacitor components are provided between the counter internal electrode layers **30**, and the capacitor components are connected in series. Therefore, the voltage applied to each capacitor component is reduced, allowing for achieving high withstand voltage of the multilayer ceramic capacitor **1**. The multilayer ceramic capacitor **1** in the present example embodiment may include a multi-portion structure including four or more portions.

[0169] In the multilayer ceramic capacitor **1** with the structures illustrated in FIGS. **7**A to **7**C, similar to the present example embodiment described above, the first external electrode **40**A includes the first main surface-side recess **510**A and the second main surface-side recess **520**A, while the second external electrode **40**B includes the first main surface-side recess **510**B and the second main surface-side recess **520**B. The surface roughness of the recesses is preferably rougher than the surface roughness of the peripheral regions of the recesses.

[0170] In the multilayer ceramic capacitor **1** with the structures illustrated in FIGS. **7**A to **7**C, similar to the present example embodiment described above, the first external electrode **40**A may include the first lateral surface-side recess **530**A and the second lateral surface-side recess **540**A, while the second external electrode **40**B may include the first lateral surface-side recess **530**B and the second lateral surface-side recess **540**B. The surface roughness of the recesses is preferably rougher than the surface roughness of the peripheral regions of the recesses.

[0171] In particular, the multilayer ceramic capacitors **1** with

[0172] the structures such as the two-portion structure, the three-portion structure, and the four-portion structure including the floating internal electrode layer **35** illustrated in FIGS. **7**A to **7**C are effective for use under high voltage. However, compressive stress is preferably applied to the multilayer body **10** as an electrostrictive countermeasure under high voltage, and in order to achieve this, the plating thickness of the external electrodes on the main surfaces and the lateral surfaces of the multilayer body **10** is increased. However, this case involves a problem of deteriorating the flexural resistance due to increased tensile stress on the inner end portion **411***e* and the inner end **560***a* of the external electrode **40** illustrated in FIG. **5**B. However, for example, as in the present example embodiment, the recesses (first main surface-side recess, second main surface-side recess) are provided to the main surface-side external electrode, and the surface roughness of the recesses is made rougher than the surface roughness of the peripheral regions, thus improving the flexural resistance, and consequently reducing or preventing the occurrence of cracks or the like in the multilayer body **10**.

[0173] The multilayer ceramic capacitor **1** according to the present example embodiment described above achieves the following advantageous effects.

[0174] The multilayer ceramic capacitor **1** according to the present example embodiment includes the plurality of dielectric layers **20** as the plurality of ceramic layers, and the plurality of internal electrode layers **30** as the plurality of inner conductive layers, both of which are stacked alternately in the lamination direction T. The multilayer ceramic capacitor **1** also includes the multilayer body **10** and the pair of external electrodes **40**. The multilayer body **10** includes the first main surface TS**1** and the second main surface TS**2** on opposite sides in the lamination direction T, the first end surface LS**1** and the second end surface LS**2** on opposite sides in the length direction L orthogonal or substantially orthogonal to the lamination direction T, and the first lateral surface WS**1** and the second lateral surface WS**2** on opposite sides in the width direction W orthogonal or substantially orthogonal to both the lamination direction T and the length direction L. The pair of external electrodes **40** are spaced apart from each other at both ends of the multilayer body **10** in the length direction L. The internal electrode layer **30** includes the first internal electrode layer **31** as the first inner conductive layer extending to the first end surface LS**1**, and the second internal electrode layer **32** as the second inner conductive layer extending to the second end surface LS**2**. The external electrodes **40** include the first main surface-side external electrode **411**A, the second main surface-side external **412**A, the first main surface-side external electrode **411**B, and the second main surface-side external electrode **412**B, as the main surface-side external electrodes provided on the first main surface TS**1** and the second main surface TS**2**, respectively. The first main surface-side external electrode **411**A, the second main surface-side external electrode **412**A, the first main surface-side external electrode **411**B, and the second main surface-side external electrode **412**B include the first main surface-side recess **510**A, the second main surface-side recess **520**A, the first main surface-side recess **510**B, and the second main surface-side recess **520**B, respectively, as the recesses recessed towards the multilayer body **10** side in the cross-sectional view along the lamination direction T and the length direction L, as well as the inner peripheral region **560** and the outer peripheral region **570** as the peripheral regions adjacent to the recess in the length direction L. The surface roughness Ra of the first main surface-side recess **510**A, the second main surface-side recess **520**A, the first main surface-side recess **510**B, and the second main surface-side recess **520**B is rougher than the surface roughness of the inner peripheral region **560** and the outer peripheral region **570**.

[0175] This enables an improvement of the flexural resistance of the multilayer ceramic capacitor **1** when mounted on a board, and consequently reduces or prevents the occurrence of cracks or the like in the multilayer body **10**.

[0176] In the multilayer ceramic capacitor **1** according to the present example embodiment, the surface roughness Ra of the first main surface-side recess **510**A, the second main surface-side recess **520**A, the first main surface-side recess **510**B, and the second main surface-side recess **520**B may preferably be between about 0.7 μm and about 1.4 μm inclusive.

[0177] This enables an improvement of the flexural resistance when mounted on a board, and consequently reduces or prevents the occurrence of cracks or the like in the multilayer body **10**.

[0178] In the multilayer ceramic capacitor **1** according to the present example embodiment, the surface roughness Ra of the inner peripheral region **560** and the surface roughness Ra of the outer peripheral region **570** may preferably be between about 0.2 μm and about 0.5 μm inclusive.

[0179] This enables an improvement of the flexural resistance when mounted on a board, and consequently reduces or prevents the occurrence of cracks or the like in the multilayer body **10**.

[0180] In the multilayer ceramic capacitor **1** according to the present example embodiment, the internal electrode layer **30** may include the floating internal electrode layer **35** as the floating inner conductive layer, which does not extend to either the first end surface LS**1** or the second end surface LS**2**, and which faces at least one of the first internal electrode layer **31** or the second internal electrode layer **32** across the dielectric layer **20**.

[0181] This achieves a high withstand voltage of the multilayer ceramic capacitor **1**.

[0182] In the multilayer ceramic capacitor **1** according to the present example embodiment, each of the first external electrode **40**A and the second external electrode **40**B may include the lateral surface-side external electrode, the lateral surface-side external electrode may also include a recess similar to that of the main surface-side external electrode, and the surface roughness of the recess may be rougher than the surface roughness of the peripheral regions.

[0183] In other words, the multilayer ceramic capacitor **1** of the present example embodiment includes the multilayer body **10**. The multilayer body **10** includes the plurality of dielectric layers **20** as the plurality of ceramic layers, and the plurality of internal electrode layers **30** as the plurality of inner conductive layers, both of which are stacked alternately in the lamination direction T. The multilayer body **10** also includes the first main surface TS**1** and the second main surface TS**2** on opposite sides in the lamination direction T, the first end surface LS**1** and the second end surface LS**2** on opposite sides in the length direction L orthogonal or substantially orthogonal to the lamination direction T, and the first lateral surface WS**1** and the second lateral surface WS**2** on opposite sides in the width direction W orthogonal or substantially orthogonal to both the lamination direction T and the length direction L. The multilayer body **10** also includes the pair of external electrodes **40** provided spaced apart from each other at both ends of the multilayer body **10** in the length direction L. The internal electrode layer **30** includes the first internal electrode layer **31** as a first inner conductive layer extending to the first end surface LS**1**, and the second internal electrode layer **32** as a second inner conductive layer extending to the second end surface LS**2**. The external electrodes **40** include the lateral surface-side external electrode provided on at least one of the first lateral surface WS**1** or the second lateral surface WS**2**. The lateral surface-side external electrode includes the recess recessed towards the multilayer body side in a cross-sectional view along the lamination direction and the length direction, and the peripheral regions adjacent to the recess in the length direction. The surface roughness of the recess is rougher than the surface roughness of the peripheral regions.

[0184] The present invention is not limited to the configurations of the example embodiments and can be appropriately modified and applied within the scope that does not change the essence of the present invention. Combinations of two or more of any individual preferable configurations described in the example embodiments are also considered a portion of the present invention.

[0185] For example, the multilayer ceramic capacitor **1** may be of a two-terminal capacitor including two external electrodes or a multi-terminal capacitor including a plurality of external electrodes.

[0186] In the example embodiments, the multilayer ceramic capacitor using dielectric ceramics has been described as an example of the multilayer ceramic electronic component. However, the multilayer ceramic electronic component disclosed herein is not limited to this, and various multilayer ceramic electronic components such as, for example, piezoelectric components using piezoelectric ceramics, thermistors using semiconductor ceramics, and inductors using magnetic ceramics can also be applied. Examples of piezoelectric ceramics may include PZT (lead zirconate

titanate) ceramics, examples of semiconductor ceramics may include spinel ceramics, and examples of magnetic ceramics may include ferrites.
[0187] While example embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. A multilayer ceramic electronic component, comprising: a multilayer body including a plurality of ceramic layers and a plurality of inner conductive layers stacked alternately in a lamination direction, first and second main surfaces on opposite sides in the lamination direction, first and second end surfaces on opposite sides in a length direction orthogonal or substantially orthogonal to the lamination direction, and first and second lateral surfaces on opposite sides in a width direction orthogonal or substantially orthogonal to both the lamination direction and the length direction; and a pair of external electrodes spaced apart from each other at both of the first and second end surfaces in the length direction; wherein the plurality of inner conductive layers include a first inner conductive layer extending to the first end surface, and a second inner conductive layer extending to the second end surface; the pair of external electrodes include a main surface-side external electrode on at least one of the first main surface or the second main surface; the main surface-side external electrode includes a recess recessed towards a multilayer body side in a cross-sectional view along the lamination direction and the length direction, and a peripheral region adjacent to the recess in the length direction; and a surface roughness of the recess is rougher than a surface roughness of the peripheral region.
2. The multilayer ceramic electronic component according to claim 1, wherein the surface roughness of the recess is between about 0.7 μm and about 1.4 μm inclusive.
3. The multilayer ceramic electronic component according to claim 2, wherein the surface roughness of the peripheral region is between about 0.2 μm and about 0.5 μm inclusive.
4. The multilayer ceramic electronic component according to claim 1, wherein the plurality of inner conductive layers include a floating inner conductive layer facing at least one of the first inner conductive layer or the second inner conductive layer across the plurality of ceramic layers, the floating inner conductive layer not extending to either of the first or second end surfaces.
5. The multilayer ceramic electronic component according to claim 1, wherein the multilayer body has a rectangular or substantially rectangular shape.
6. The multilayer ceramic electronic component according to claim 1, wherein a dimension of the multilayer body in the length direction is between about 0.2 mm and about 10 mm inclusive; a dimension of the multilayer body in the lamination direction is between about 0.05 mm and about 10 mm inclusive; and; a dimension of the multilayer body in the width direction is between about 0.1 mm and about 10 mm inclusive.
7. The multilayer ceramic electronic component according to claim 1, wherein each of the plurality of ceramic layers includes at least one of BaTiO.sub.3, CaTiO.sub.3, SrTiO.sub.3, or CaZrO.sub.3 as a main component.
8. The multilayer ceramic electronic component according to claim 7, wherein each of the plurality of ceramic layers includes at least one of Mn compounds, Fe compounds, Cr compounds, Co compounds, or Ni compounds.
9. The multilayer ceramic electronic component according to claim 1, wherein a thickness of each of the plurality of ceramic layers is between about 0.2 μm and about 15 μm inclusive.
10. The multilayer ceramic electronic component according to claim 1, wherein a number of the plurality of ceramic layers is between 10 and 1200 inclusive.
11. The multilayer ceramic electronic component according to claim 1, wherein each of the

plurality of inner conductive layers includes at least one of Ni, Cu, Ag, Pd, or Au, or an alloy including at least one of Ni, Cu, Ag, Pd, or Au.

12. The multilayer ceramic electronic component according to claim 1, wherein a thickness of each of the plurality of inner conductive layers is between about 0.2 μm and about 2.0 μm inclusive.

13. The multilayer ceramic electronic component according to claim 1, wherein a number of the plurality of inner conductive layers is between 10 and 1000 inclusive.

14. The multilayer ceramic electronic component according to claim 1, wherein each of the pair of external electrodes includes a base electrode layer and a plated layer on the base electrode layer.

15. The multilayer ceramic electronic component according to claim 14, wherein the base electrode layer is a fired layer including a metal component and at least one of a glass component and a ceramic component.

16. The multilayer ceramic electronic component according to claim 15, wherein the metal component includes at least one of Cu, Ni, Ag, Pd, Ag—Pd alloy, or Au.

17. The multilayer ceramic electronic component according to claim 15, wherein the glass component includes at least one of B, Si, Ba, Mg, Al, or Li.

18. The multilayer ceramic electronic component according to claim 15, wherein the ceramic component includes at least one of BaTiO.sub.3, CaTiO.sub.3, (Ba, Ca) TiO.sub.3, SrTiO.sub.3, or CaZrO.sub.3.

19. The multilayer ceramic electronic component according to claim 14, wherein plated layer includes a Ni plated layer on the base electrode layer and a Sn plated layer on the Ni plated layer.